Indian Institute of Information Technology Vadodara

CS/IT 429

Embedded Systems

Dr. Yash Agrawal
Visiting Faculty, IIIT Vadodara
Associate Professor, DA-IICT Gandhinagar

Embedded Systems

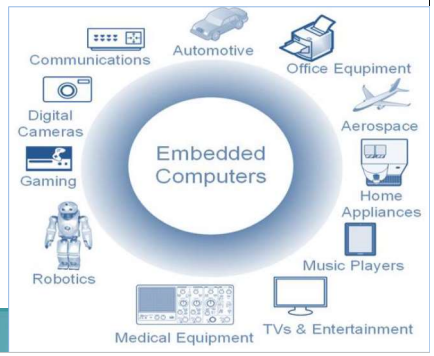
What is General Purpose Computing Systems..?

- Able to run variety of software.
- Contains relatively high-performance hardware components (fast processors, data and program storage).
- Requires an operating system.
- Uses a variety of peripherals (displays, keyboard, mice, internet connections, wireless communication capability, etc.)
- It can be used for multiple applications such as professional presentation, report making, entertainment – video, image displaying, gaming, etc.

Embedded Systems

What is Embedded System?

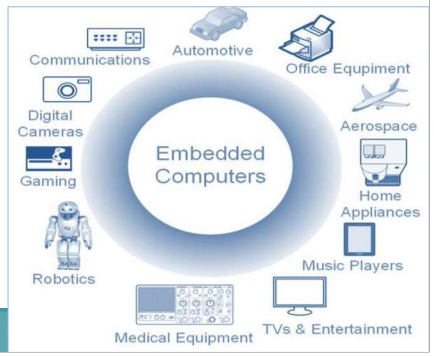
- Resources can be implemented on a single IC.
- Include a variety of peripherals (timers, A-D converters, D-A converters, interfaces).
- Small Size makes them very versatile.
- Uses a variety of peripherals (displays, keyboard, mice, internet connections, wireless communication capability, etc.)



Embedded Systems

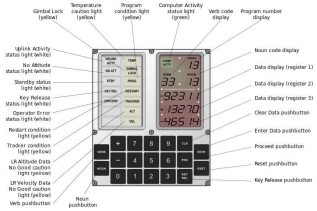
What is Embedded System?

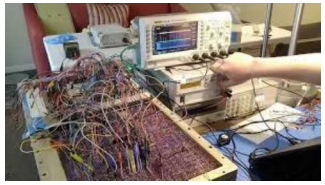
- Contains firmware (only the needed software which is not intended to be changed frequently).
- May contain Real Time Operating Systems (RTOS) which are used as a task scheduler..
- Relatively lesser costly.

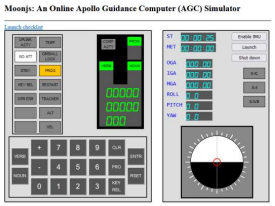


History of Embedded Systems

Apollo Guidance Computer is considered as one of the earliest systems with all components of ES. It was developed at MIT Instrumentation Laboratory by Charles Stark Draper in the early 1960s. The system contained thousands of integrated circuit (IC) chips (4100 for Block I and 2800 for Block II) with NOR gates performing the computations with word length of 16-bits. The system had an integrated interface with display and keyboard (DSKY). The software (instruction set) was custom developed assembly program with 3 bits for opcode and 12 bits for address. The Block I design had 12 kilowords of fixed memory (later updated to 24 kilowords), and the Block II design had 32 kilowords of fixed memory and 4 kilowords of erasable memory. The system used a 2.048 MHz crystal clock kilowords of erasable memory. The system used a 2.048 MHz crystal clock. The ES technology has come quite a long way since then.







[Online Simulator: https://svtsim.com/moonjs/agc.html]

[Embedded Systems – A Hardware-Software co-design approach: Bashir I. Morshed]

History of Embedded Systems

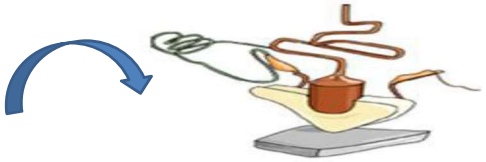
Embedded Systems has progressed drastically in higher computation capability and lower cost with the advent of microcontrollers, higher memory capacity, and increased clock speed. In the latter half of 1960s, complex ICs were developed for high-volume use and companies like Intel was formed for commercial production of ICs. The trend in technological growth tends to follow Moore's Law predicted in 1965 by Intel co-founder Gordon Moore. The law states that IC transistor capacity will double roughly every 18 months, and the trend is observed for the past several decades.

By the early 1980s, microcontrollers have integrated input/output system components as well as memory into the same chip as the processor. The size and weight of newer devices are becoming smaller with the same or better functionalities, while the comfort and unobtrusiveness are increasing at a rapid pace

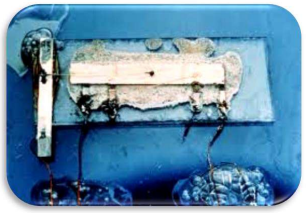
History of Embedded Systems

Solid state devices and phenomenon became popular with the invention of transistors in 1947.

Vacuum tube, 1906



Transistor, 1947



Integrated circuit, 1958

System Development

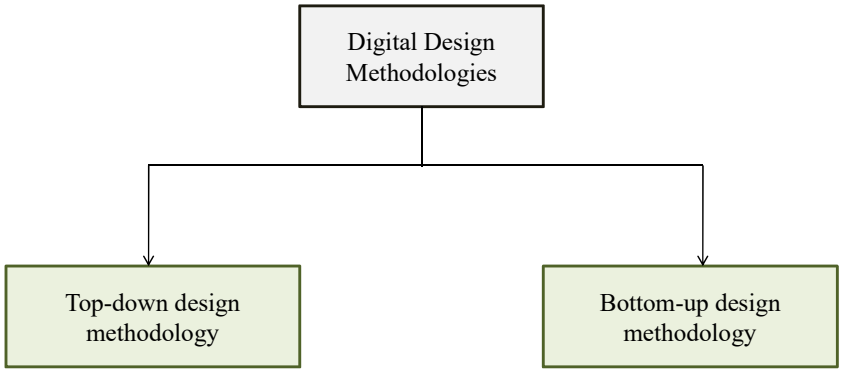
Systems

Types of Systems

Contd…

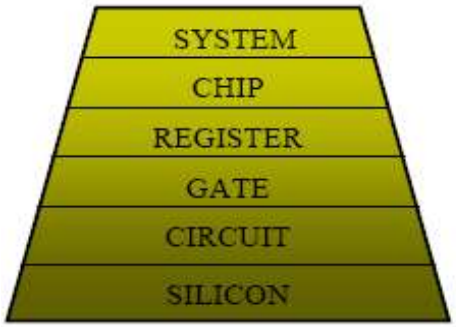
Basic Design Flow

Design, Methodologies, Hierarchy, Levels of Abstraction



Levels of abstraction and tools used…

Design Abstraction Levels



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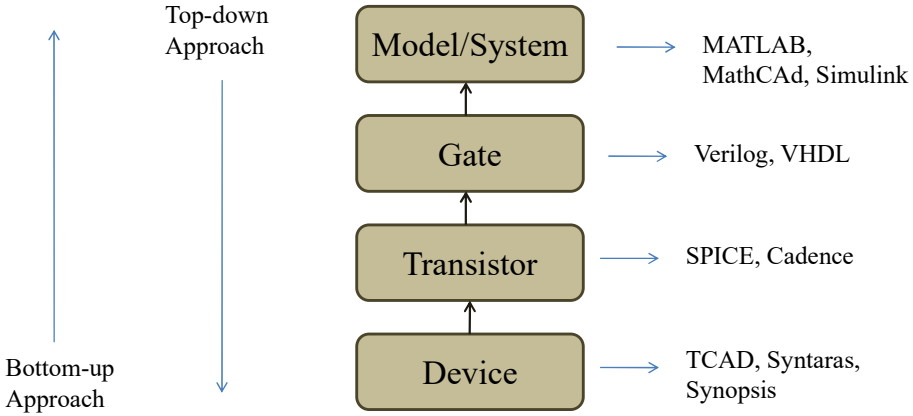
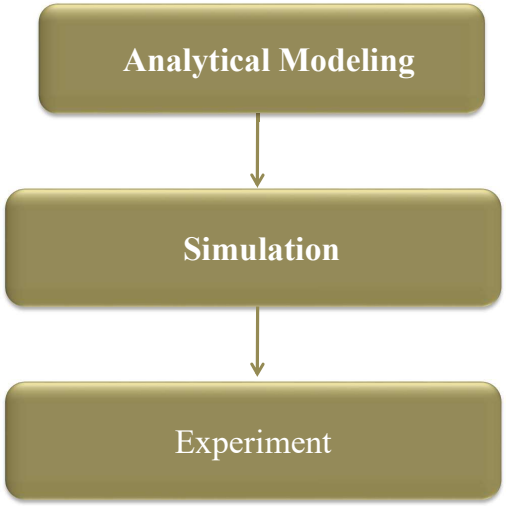


Abstraction Levels to Solve Engineering Designs and Systems



Contd….

Analytical Modeling Techniques

Closed form

Numerical form

Contd….

Simulation Techniques/Platforms

Atomic Level

Device Level

Circuit Level

Gate Level

System Level

Embedded Systems

Experimental Hardware Abstraction Levels

Bread board

Printed Circuit Board

Microprocessor and Microcontrollers (8051, ATMega, ARM…)

Development Boards -- Arduino, Rapberry Pi, etc.

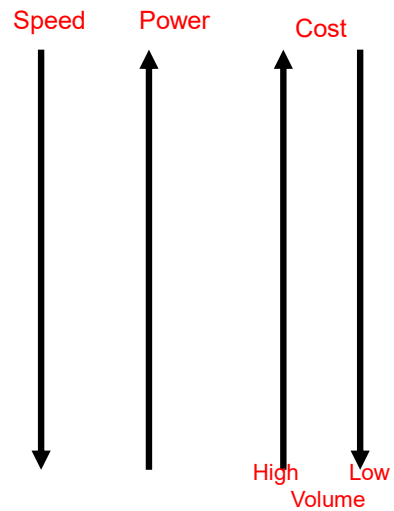
Programmable Devices (PROM, PAL, PLA, CPLD, FPGA)

Application Specific Integrated Circuit

Embedded Systems

Many Implementation Choices

- Microprocessors
- Domain-specific processors
  - DSP
  - Network processors
  - Microcontrollers
- ASIPs
- Reconfigurable SoC
- FPGA
- GateArray
- ASIC
- Full-Custom (COTS)



Slide courtesy of Mani Srivastava®

Applications of VLSI & ES

Some examples of Embedded systems

Anti-lock brakes
Auto-focus cameras
Automatic teller machines
Automatic toll systems
Automatic transmission
Avionic systems
Battery chargers
Camcorders
Cell phones
Cell-phone base stations
Cordless phones
Cruise control
Curbside check-in systems
Digital cameras
Disk drives
Electronic card readers
Electronic instruments
Electronic toys/games
Factory control
Fax machines
Fingerprint identifiers
Home security systems
Life-support systems
Medical testing systems
Modems
MPEG decoders
Network cards
Network switches/routers
On-board navigation
Pagers
Photocopiers
Point-of-sale systems
Portable video games
Printers
Satellite phones
Scanners
Smart ovens/dishwashers
Speech recognizers
Stereo systems
Teleconferencing systems
Televisions
Temperature controllers
Theft tracking systems
TV set-top boxes
VCR's, DVD players
Video game consoles
Video phones
Washers and dryers
And so on……

[Setha Pan-ngum]

Applications of Embedded Systems

- On average electronic-gadgets in home have around 40 to 50 embedded processors in it
  - Microwave, washer, dryer, dishwasher, TV, VCR, stereo, hair dryer, coffee maker, remote control, humidifier, heater, toys, etc.
- Luxury cars have over 60 embedded processors
  - Brakes, steering, windows, locks, ignition, dashboard displays, transmission, mirrors, etc.
- Personal computers have over 10 embedded processors
  - Graphics accelerator, mouse, keyboard, hard-drive, CD-ROM, bus interface, network card, etc.

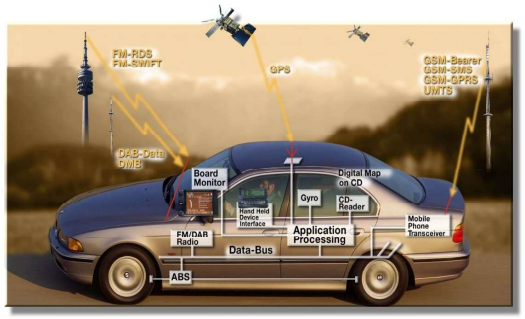
[Mike Schulte]

Applications of Embedded Systems

Embedded Automotive

- Sensors and actuators are distributed all over the vehicle
- Functions by embedded processing:
  - ABS: Anti-lock braking systems
  - ESP: Electronic stability control
  - Airbags
  - Efficient automatic gearboxes
  - Theft prevention with smart keys
  - Blind-angle alert systems, etc ...
- Multiple Processors ~ 100

  Large diversity in processor types:
  - 8-bit – door locks, lights, etc.
  - 16-bit – most functions
  - 32-bit – engine control, airbags



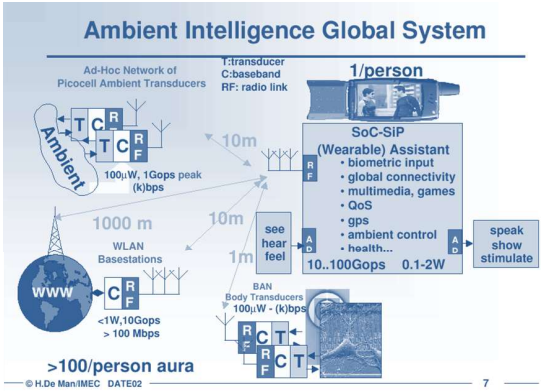
- More than 30% of the cost of a car is now in Electronics
- 90% of all innovations will be based on electronic systems

[Alberto Sangiovanni-Vincentelli®]

Applications of Embedded Systems

Consumer Electronics

- **Consumer electronics:** cellular phones, settop boxes, televisions, remote controls, game consoles, Internet appliances, PDAs, Alarm Systems, hi-fi systems, home cinemas,…
- **Home appliances** ("White Appliances") like refrigerators, washing machines (…which now-days include microprocessors and may also have internet connection…)



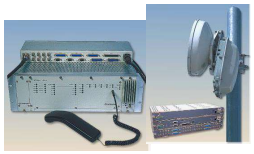
Applications of Embedded Systems

Telecommunications

-- Wireline Access Systems (copper enhancement, DSLAM)
-- Wireless Access Systems (microwave systems)
-- Terminal Equipments (ISDN modems, DSL modems, IP phones, payphones, etc.)

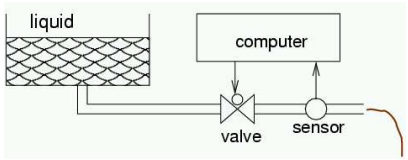
Defense Systems

-- Secure Communications
-- Crypto-systems



Applications of Embedded Systems

Fabrication Equipment

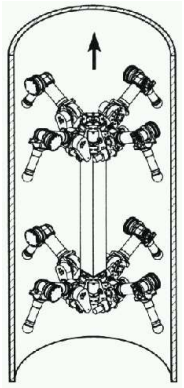


Smart Buildings

Show movie http://www.date-conference.com/conference/2003/keynotes/index.htm

Applications of Embedded Systems

Robotics



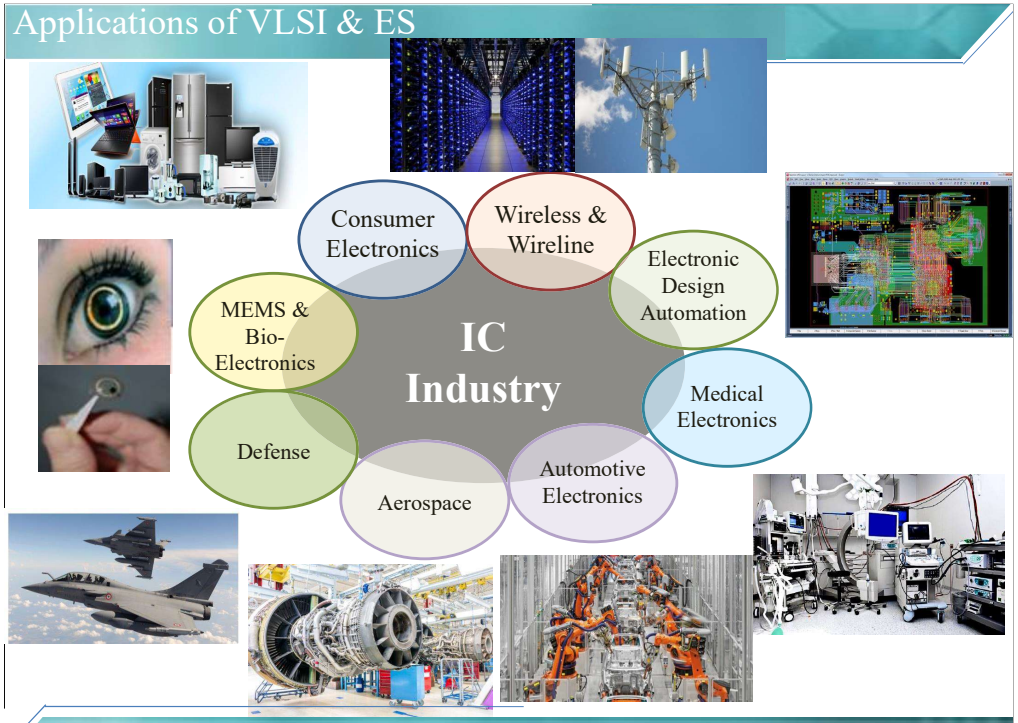
[Robot „Johnnie" (Courtesy and ©: H.Ulbrich, F. Pfeiffer, TU München]

Show movie of 2-legged robot(s)

IC Products

- Processors
  - CPU, DSP, Controllers
- Memory chips
  - RAM, ROM, EEPROM
- Analog
  - Mobile communication, audio/video processing
- Programmable
  - PLA, FPGA
- Embedded systems
  - Used in cars, factories
  - Network cards
- System-on-chip (SoC)

Applications of VLSI & ES



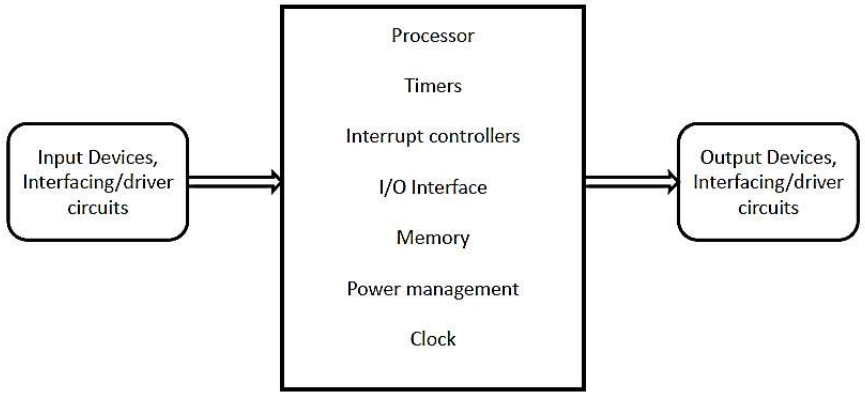
VLSI and Embedded Systems….

Technology Advancement and Need of advanced **Applications…**

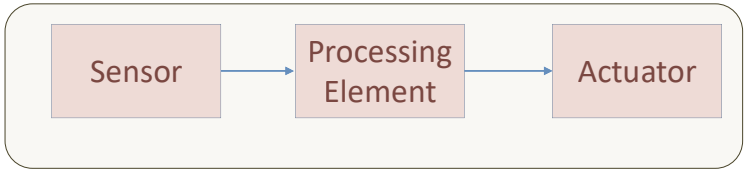
Embedded Systems	SSI, MSI, LSI, VLSI	Embedded Systems	VLSI, ULSI	Embedded Systems
Connecting discrete active and passive Components, transistor on PCB, Bread board	Integrated Circuit (IC) – Integrating millions of Transistors	Embedding and Connecting Several ICS, Components, Sensors and Actuators	Again Integrating discrete Components of Embedded Systems to develop entire System or Network on a Chip	Further Embedding of Systems can be made to get advanced Applications
Printed Circuit Board (PCB), Bread board	Integrated Circuit (IC)	Internet of Things (IOT) Cyber Physical **Systems**	System-on-Chip (SoC) Network-on-Chip (NoC)	

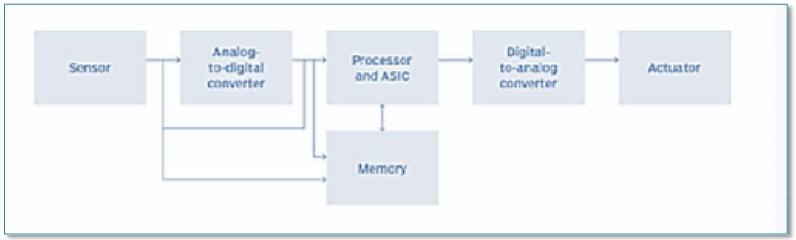
Embedded Systems

Block-level representation of Embedded System

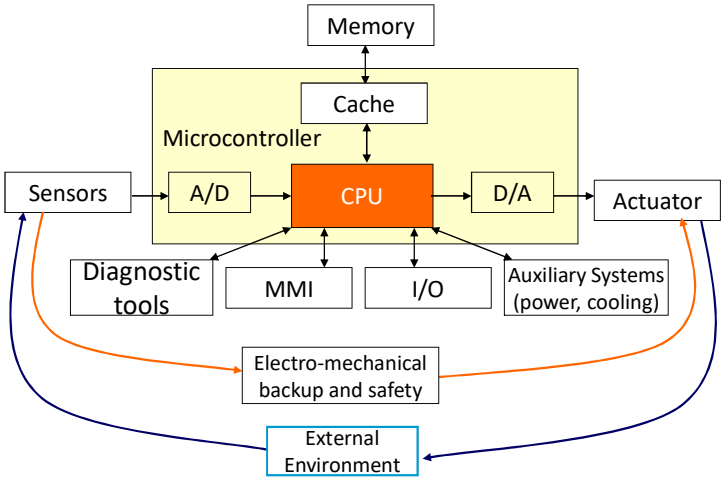


Generic Structure of Embedded System

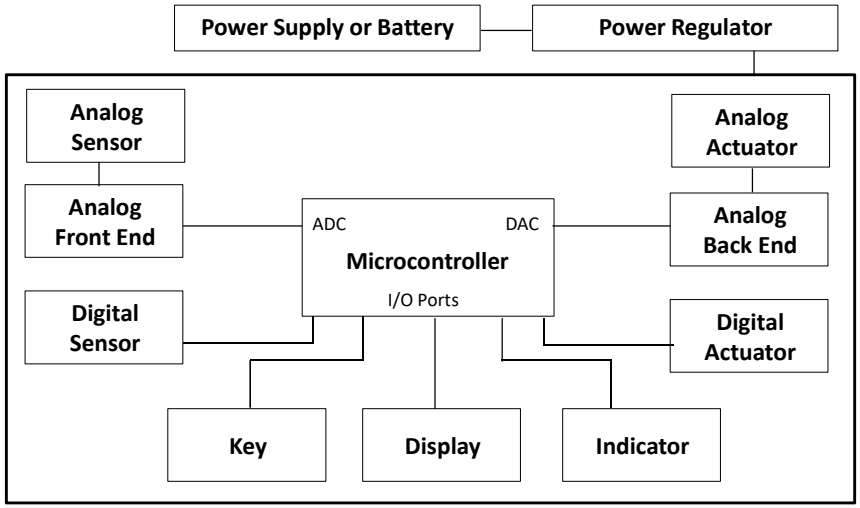




Generic Structure of Embedded System

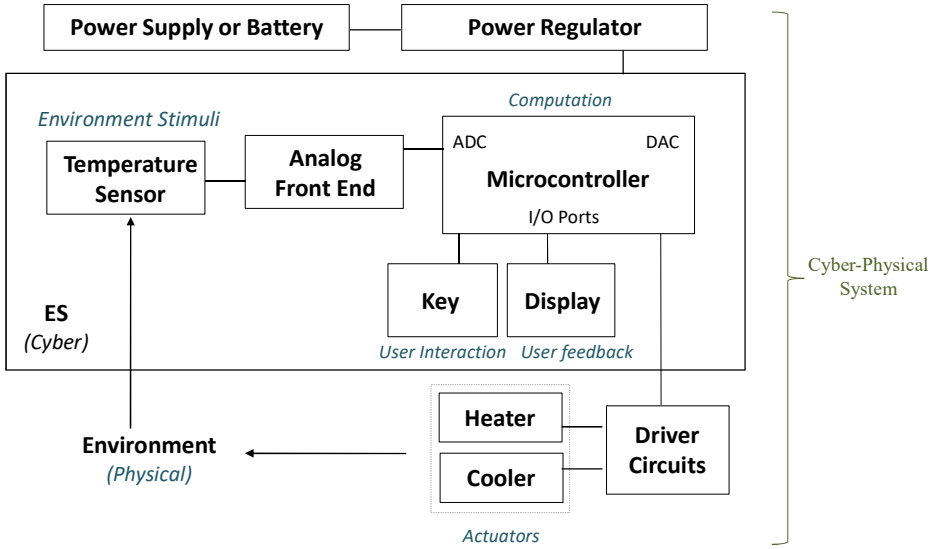


Typical Hardware Composition of an Embedded System



[Embedded Systems – A Hardware-Software co-design approach: Bashir I. Morshed]

Example: Can you design the block-level design to implement air conditioning system..?



Embedded Systems

Components of Embedded Systems Hardware

- **Microprocessor (µP):** It has CPU (control unit and ALU), caches, floating point ALU, pipelining etc. µP is used when large embedded software is to be located in the external memory
- **Microcontroller (µC):** It is single VLSI chip having limited computational capabilities, it has enhanced I/O capabilities and number of on-chip functional unit. µC Is used when a small part of the embedded software has to be located in the internal memory and when on-chip functional unit is needed
- **Digital signal Processor (DSP):** DSP is a specialized microprocessor with its architecture optimized for the operational needs of digital signal processing. The goal of DSP is usually to measure, filter or compress continuous real-world analog signals.

Embedded Systems

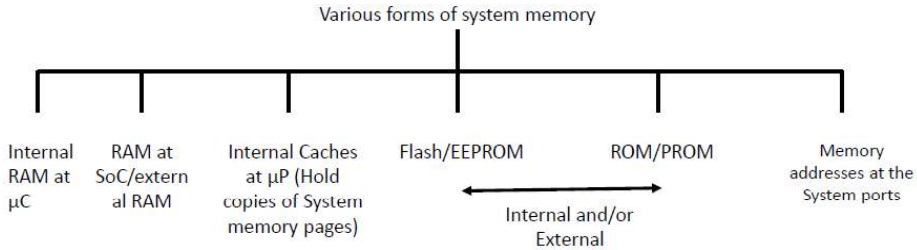
Components of Embedded Systems Hardware

- **Power Management:** This unit supply required voltage and current to the processor
- **Clock:** It controls the various clocking requirements of the CPU, of the system timers and CPU machine cycles. Clock control the time for executing an instruction
- **Interrupt handler:** An interrupt handler must exist in all embedded systems to handle the interrupt from various process in the system.
- **I/O interface:** An embedded system get inputs from the physical devices (using read operations at port addresses), process the input and send the output (using wrote operation to port address). A system connects to eternal physical devices through parallel or serial I/O ports. A system interacts with the other devices through an I/O bus (I^2C, SPI, USB, PCI etc)
- **Timers:** A timer is a specialized type of clock which is used to measure time intervals. It is a device that counts down from a specified time interval and used to generate a time delay.
- **Counters:** A counter is a unit that stores the number of times a particular event or process has occurred, with respect to a clock signal

Embedded Systems

Components of Embedded Systems Hardware

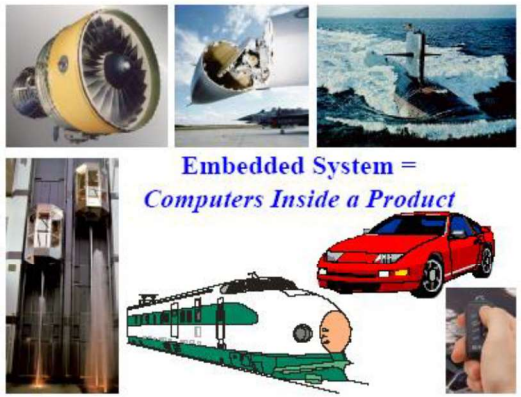
- **Memories**



Embedded Systems

Entities in the Embedded Systems

- Hardware
- Software
- Mechanical Components



[P Koopman, CMU]

Example

Example 1: Washing Machine:
A washing machine from an embedded systems point of view has:

a. Hardware: Buttons, Display & buzzer, electronic circuitry.
b. Software: It has a chip on the circuit that holds the software which drives controls & monitors the various operations possible.
c. Mechanical Components: the internals of a washing machine which actually wash the clothes control the input and output of water, the chassis itself.

Example 2: Air Conditioner
An Air Conditioner from an embedded systems point of view has:

a. Hardware: Remote, Display & buzzer, Infrared Sensors, electronic circuitry.
b. Software: It has a chip on the circuit that holds the software which drives controls & monitors the various operations possible. The software monitors the external temperature through the sensors and then releases the coolant or suppresses it.
c. Mechanical Components: the internals of an air conditioner the motor, the chassis, the outlet, etc.

Characteristics of Embedded Systems

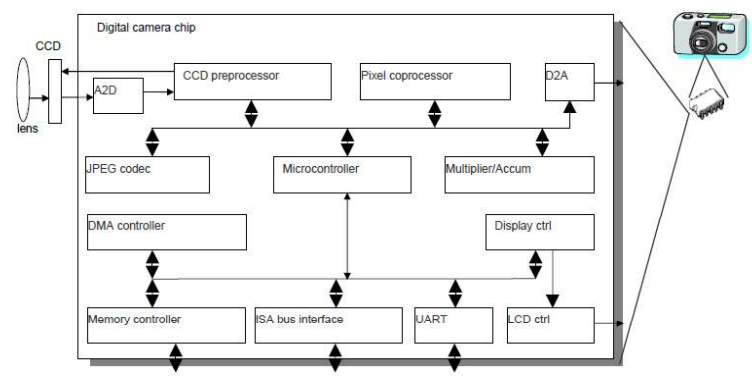
- An electronic device that incorporates a processing unit such as microcontroller.
- It contains hardware and software needed to operate the electronic device.
- It is designed and can operate independently without human intervention.
- The device is optimized to handle a few specific tasks only (as per the design).

[D. W. Lewis: Fundamentals of Embedded Software with the ARM Cortes-M3, Pearson Education]

Features of Embedded Systems

- Function Specific
- Tightly Constraint
- Real-time Reactive Operation
- Low Cost
- Low Power

An embedded system example – digital camera



- Single-functioned -- always a digital camera
- Tightly-constrained -- Low cost, low power, small, fast
- Reactive and real-time -- only to a small extent

Embedded Systems

Factors that should be considered at design time

- Efficient
- Reliable
- Maintainable
- Safe
- Secure
- Real-time

Classification of Embedded Systems

Some Classification Approaches to Embedded Systems

- Generation
- Complexity and Performance
- Deterministic Behaviour
- Triggering
- Generic Applications

Classification of Embedded Systems

Classification of Embedded Systems can be based on Generation

- First generation(1G):
  - Built around 8bit microprocessor & microcontroller.
  - Simple in hardware circuit & firmware developed.
  - Examples: Digital telephone keypads.
- Second generation(2G):
  - Built around 16-bit µp & 8-bit µc.
  - They are more complex & powerful than 1G µp & µc.
  - Examples: SCADA systems
- Third generation(3G):
  - Built around 32-bit µp & 16-bit µc.
  - Concepts like Digital Signal Processors(DSPs),
  - Application Specific Integrated Circuits (ASICs) evolved.
  - Examples: Robotics, Media, etc.
- Fourth generation:
  - Built around 64-bit µp & 32-bit µc.
  - The concept of System on Chips (SoC), Multicore
  - Processors evolved.
  - Highly complex & very powerful.
  - Examples: Smart Phones.

[….]

Classification of Embedded Systems

Classification of Embedded Systems can be based on Complexity and Performance

- Small-scale:
  - Simple in application need
  - Performance not time-critical
  - Built around low performance & low cost 8 or 16 bit µp/µc
  - Example: an electronic toy
- Medium-scale:
  - Slightly complex in hardware & firmware requirement
  - Built around medium performance & low cost 16 or 32 bit µp/µc
  - Usually contain operating system
  - Examples: Industrial machines
- Large-scale:
  - Highly complex hardware & firmware
  - Built around 32 or 64 bit RISC µp/µc or PLDs or Multicore Processors
  - Response is time-critical
  - Examples: Mission critical applications

[….]

Classification of Embedded Systems

Classification of Embedded Systems can be based on Deterministic Behaviour

This classification is applicable for "Real Time" systems.

- The task execution behaviour for an embedded system may be deterministic or non-deterministic.
- Based on execution behaviour Real Time embedded systems are divided into Hard and Soft.

Hard Real time system

A real-time operating system should strictly adhere to the timing constraints for a task. A hard real-time system must meet the timing deadlines without any delay. Missing the deadline would cause serious failure to the system or user.

The airbag control system and antilock braking system of vehicles are typical examples for hard real-time systems. When a vehicle is met with an accident, the airbag control system should operate immediately without any delay to safeguard the passenger. If there is any delay in the deployment of airbags, it will lead to the death of passengers in the vehicle.

So for a hard real-time system, "*A late answer is always a wrong answer*".

Soft Real time system

In these types of embedded systems deadline is not strictly followed. Missing deadlines for tasks are acceptable for soft real-time systems, but the frequency of deadlines missing should be within the compliance limit.

[….]

Classification of Embedded Systems

Classification of Embedded Systems can be based on Triggering

Embedded systems which are "Reactive" in nature can be based on triggering. Reactive systems can be:

- Event Triggered
- Time Triggered

The embedded systems are classified into two types based on the triggering of the systems: time-triggered and event-triggered. If a system is activated or triggered based on the pre-defined task or preset time, then such a system is said to be the time-triggered embedded system.

On the other side, if the system is triggered based on some activity like change in temperature or change in pressure, such system is said to be an event triggered embedded system.

[….]

Embedded Systems Categories

Classification of Embedded Systems can be based on Generic Applications

- General Computing
  - Applications similar to desktop computing, but in an embedded package (lower cost, targeting also computer illiterates, etc.)
  - Set-top boxes, game consoles, internet appliances, automatic tellers, wearable computers), etc.
- Control Systems
  - Closed-loop feedback control of real-time systems
  - Vehicle engines, chemical process, industrial process, nuclear power, flight control, etc.
- Signal Processing
  - Computations involving large data streams (signals)
  - Radar, Sonar, video compression, Digital TV / HDTV broadcast centers, etc.
- Communication & Networking
  - Switching and information transmission
  - Telephone systems, Internet, satellite systems, etc.

[….]

Design Metrics

Some Common Metrics

- Unit cost: the monetary cost of manufacturing each copy of the system, excluding NRE cost
- NRE cost (Non-Recurring Engineering cost): The one-time monetary cost of designing the system
- Size: the physical space required by the system
- Performance: the execution time or throughput of the system
- Power: the amount of power consumed by the system

[Vahid/Givargis, Embedded Systems Design: A Unified Hardware/Software Introduction]

Design Metrics

Some Common Metrics

- Flexibility: the ability to change the functionality of the system without incurring heavy NRE cost
- Time-to-prototype: the time needed to build a working version of the system
- Time-to-market: the time required to develop a system to the point that it can be released and sold to customers
  Time required to develop a product from concept development to the point that it can be sold to the customers.
- Maintainability: the ability to modify the system after its initial release
- Correctness, accuracy, precision, safety, reliability, redundancy, many more

[Vahid/Givargis, Embedded Systems Design: A Unified Hardware/Software Introduction]

Design Metrics

Other Design Constraints

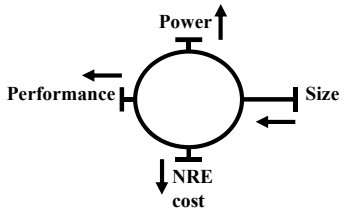
- Dependability
  - Reliability: probability of system working correctly provided that it worked at time t=0
  - Maintainability: probability of system working correctly d time units after error occurred. [Some systems require no maintenance throughout their operating lives (e.g. electric kettles, computer keyboards), while some may need it such as mobile phones and airplane flight control (software upgrade)]
- Operating environment
  Some engine Electronic Control Units (ECUs) in cars are located under the bonnets. So they have to work at high temperature, as well as dusty and wet environment.
- EMI (Electromagnetic Interference)

[Setha Pan-ngum]

Design Challenges

Trade-off in Design Metrics: Improving one may worsen others

- Optimizing design metrics is a key challenge of ES designers.
- Expertise with both software and hardware is needed to optimize design metrics
  - Not just a hardware or software expert, as is common
  - A designer must be comfortable with various technologies in order to choose the best for a given application and constraints



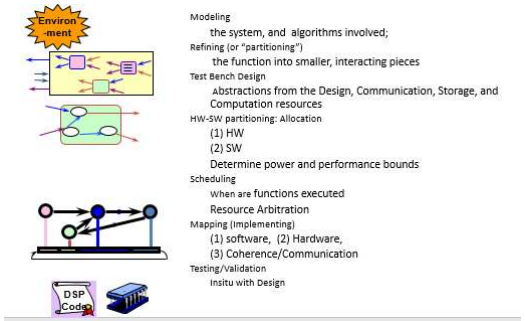
[Vahid/Givargis, Embedded Systems Design: A Unified Hardware/Software Introduction]

Embedded Systems

Hardware-Software Co-design

- Many functions can be done by **software** on a general purpose microprocessor OR by **hardware** on an application specific ICs (ASICs)
- For examples: game console graphic, PWM, PID control
- Leads to Hardware/Software Co-design concept

Embedded System Design Flow

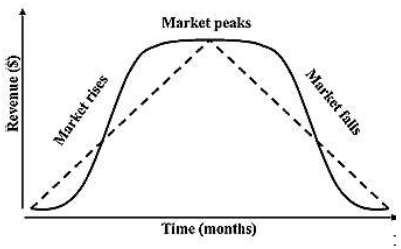


Market-size and Importance of Embedded Systems

- Embedded Systems dominate the world-wide computer system market.
- The market of Embedded System is billion of US dollars.
- While Desktop PC market is saturating, embedded market is growing.
- Embedded Systems accounts for 90-95% of all microprocessors produced worldwide!

Embedded System Market

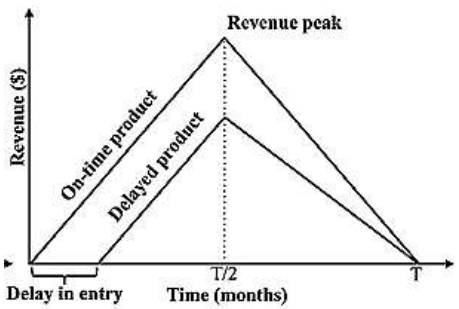
Time to Market Strategy: A demanding design metric



- Time to Market: Time required to develop a product from concept development to the point that it can be sold to the customers.
  Time to market can be as small as 6 months!
- Figure (left) shows Market window curve where marker demand starts to grow leading to increase of revenue. At some point market saturates and product revenue peaks. After some time, market demand starts to fall.
- The dotted line shows a triangular approximation of the curve for ease of analysis.

Embedded System Market

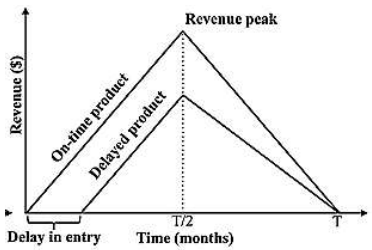
Time to Market Strategy: A demanding design metric



- Simplified revenue model
  Let,
  T be the total product life;
  P denotes the peak revenue and
  R defines the total revenue.
  The area under the curve determines the total revenue generated from the product
- Loss
  - The difference between the on-time and delayed triangle areas
  - Delays can be costly

Embedded System Market

Time to Market Strategy: A demanding design metric



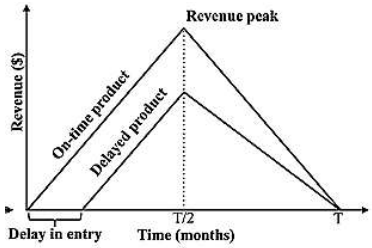
- On-Time Product

  The area under the curve determines the total revenue generated from the product

  Area = 1/2 * base * height

  => Revenue Generated $R = \frac{1}{2} T * P$

Time to Market Strategy: A demanding design metric



- Product with Delayed Entry (by D months)

Assuming the same rate of market penetration rate as the on-time product, the peak revenue of the delayed product is

$$P' = \frac{2P}{T}\left(\frac{T}{2} - D\right)$$

Hence, the total revenue of the delayed product can be given as:

$$R' = \frac{P}{T}\left(\frac{T}{2} - D\right)\ T - D$$

Potential revenue loss can be defined as the difference between the total revenue generated by the on-time product and the total revenue generated by the delayed product.

$$\%\text{Revenue Loss} = \frac{R - R'}{R} x100\% - \frac{3TD - 2D^2}{T^2} x100\%$$

Example

- A Product has life cycle of 52 weeks. Compute the percentage loss for a product delayed by 4 weeks.

  Ans: T = 52 weeks, D = 4 weeks, % Loss of Revenue ≈ 21%

- Consider the same product with life cycle of 52 weeks. However, another product is now delayed by 10 weeks. Compute the percentage loss of revenue in this case.

  Ans: T = 52 weeks, D = 10 weeks, % Loss of Revenue ≈ 50%

Inference: The second product got delayed by one-fifth of the product life-cycle has lost half of the revenue. Thus, product delay can be highly costly…

Per Unit Cost

Per unit cost is another important market factor to consider for Embedded Systems product. It can be defined as:

$$\text{Per unit cost} = \text{Unit cost} + \frac{\text{NRE cost}}{\text{Number of units}}$$

$$\text{Total cost} = \text{NRE cost} + \text{Unit cost} \bullet \text{Number of units}$$

Per Unit Cost

Example

- A product has NRE cost as Rs. 2000 and unit cost is Rs. 100.
  Compute the per-unit cost if the units produced are
  (a)10 units
  (b)1000 units

  Ans: NRE cost = Rs. 2000, Unit cost = Rs. 100
  (a) Per-unit cost = Rs. 100 + Rs. 2000/10 = Rs. 300
  (b) Per-unit cost = Rs. 100 + Rs. 2000/1000 = Rs. 102

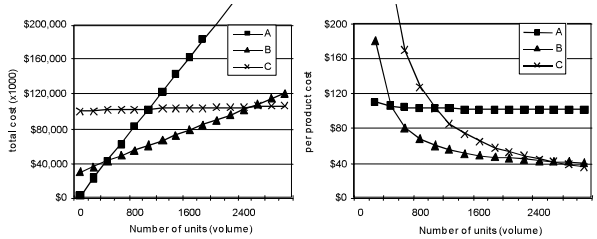
Inference: Large volume production can lead to lower per-unit cost of the product.

Embedded System Market

Per Unit Cost

Example

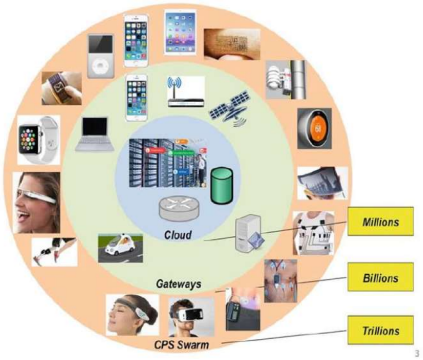
- Compare technologies by costs -- best depends on quantity
  - Technology A: NRE=$2,000, unit=$100
  - Technology B: NRE=$30,000, unit=$30
  - Technology C: NRE=$100,000, unit=$2



- But, must also consider time-to-market

Future Trends

Currently, billions of ES units are produced yearly versus millions of desktop units. It is projected that by 2030, there will be trillions of ES systems and edge nodes worldwide connected to billions of ES gateways to communicate with cloud in a massive interconnected system. Tremendous growth and usage of ES is forecasted for future.



Vision 2030 for ES forming a cyber-physical system (CPS) swarm

[Embedded Systems – A Hardware-Software co-design approach: Bashir I. Morshed]

Future Trends

Some impactful emerging sectors that ES is predicted to influence are:

- Internet of Things (IoT)
- Wearables
- Electronic Medical Devices
- Artificial Implants, Limbs and Organs
- Autonomous Vehicles
- Smart Homes, Offices, Buildings
- Cyber-Physical Systems (CPS)
- Robots and Humanoids
- Drones
- Smart Power Grid

[Embedded Systems – A Hardware-Software co-design approach: Bashir I. Morshed]

I am available/approachable at

email: yash_agrawal@iiitvadodara.ac.in
yash_agrawal@daiict.ac.in